DE–477 **11**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2014.

First Semester

PRINCIPLES OF INFORMATION TECHNOLOGY

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the data storage hierarchy and why the key field is important. (10)

   (b) What is copyright? Explain about piracy plagiarism and ownership of images and sounds. (10)

2. (a) Discuss about the ethics of information technology. (10)

   (b) Explain the four types of application software in detail. (10)

3. (a) Explain the electronic mail software and internet web browsers in detail. (10)

   (b) Write a brief note on word processing software. (10)

4. (a) Explain hard disks and optical disks in detail. (10)

   (b) Write in detail about online information services. (10)

5. (a) What are the internet, its connections, addresses and its features? (10)

   (b) Write in detail about telecommuting, virtual offies and mobile work places. (10)

6. (a) What is local networks? Explain types, components, topologies and impact of LAN's. (10)

   (b) What are the factors affecting data transmission? (10)

7. (a) Explain the types of wired and wireless channels and types of wireless communications. (10)

   (b) What are the ethics concerns about the uses of databases? (10)

8. (a) Explain the six computer based information systems and its purposes. (10)

   (b) Explain six key security issues for information technology. (10)

DE–478 **12**

DISTANCE EDUCATION

M.C.A. (S). DEGREE EXAMINATION, DECEMBER 2014.

First Semester

DIGITAL COMPUTER ORGANIZATION

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain any four number conversion techniques from one base to another base. (10)

   (b) Write a note on number codes and character codes. (10)

2. (a) Simplify the Boolean function

   $F = (W, X, Y, Z) = \sum(0,1,2,4,5,6,8,9,12,13,14)$ using four variable Karnaugh map. (10)

   (b) Explain the multiplexer and demultiplexer with logic diagrams. (10)

3. (a) Describe the construction of half adder and full adder with functional table. (10)

   (b) Describe the implementation of a combined addition and subtraction in a parallel arithmetic element. (10)

4. (a) Explain the four bit parallel binary adder in detail. (10)

   (b) Explain the master-slave flip-flop with its functional table. (10)

5. (a) Describe the binary counters in detail with its wave forms. (10)

   (b) Discuss about the magnetic tape memories and magnetic disk memories. (10)

6. (a) Describe the dynamic random access memory with pin diagram. (10)

   (b) Explain the linear select memory organization. (10)

7. (a) Explain the controlling arithmetic operations in detail. (10)

   (b) Discuss about sequence of operation of control registers. (10)

8. (a) Write in detail about pheripheral devices. (10)

   (b) Discuss about programmed I/O and instruction word. (10)

DE–479 **13**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2014.

First Semester

C AND DATA STRUCTURES

(2005 to 2010 Calender year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. (a) Write in detail the use of 'break', continue and 'goto' statement in C with the help of a program. (10)

   (b) Write short notes on : (10)

   (i) Identifiers and keywords

   (ii) Constants and variables.

2. (a) How can input and output be done in C? Explain with a C program. (10)

   (b) Write a note on looping using 'FOR' statement in C with the help of a C program. (10)

3. (a) What is recursion? Explain with a C program. (10)

   (b) How can we pass arrays to functions in 'C'? Explain with a C program. (10)

4. (a) What are strings? Write a program in C to find and replace a given substring. (10)

   (b) What is an array of pointers? Explain with a program. (10)

5. (a) How can a data file be created? Explain with a C program. (10)

   (b) How can we pass functions to other functions? (10)

6. (a) Explain stack structure and its operations. (10)

   (b) What is a Queue? Explain its operations. (10)

7. (a) What is a singly linked list? Explain. (10)

   (b) What is a binary tree? Explain. (10)

8. (a) How can lists be represented using binary trees? Explain. (10)

   (b) What is a doubly linked list? Explain. (10)

DE–480 **14**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2014

First Semester

OFFICE AUTOMATION

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss about working with multiple windows environment. (10)

   (b) Explain how to create a new document in MS-Word. (10)

2. (a) Explain the help option in MS-office. (10)

   (b) Write notes on editing features in MS-word. (10)

3. (a) Explain the mail merge features of MS-word with an example. (10)

   (b) Discuss the features in format menu in detail. (10)

4. (a) Explain Edit, Delete, Move and Copy data in Excel. (10)

   (b) Explain find and replace data option in Excel. (10)

5. (a) Explain the method of creating charts in Excel. (10)

   (b) Explain conditional formatting with an example. (10)

6. (a) Explain the procedure of creating and editing Powerpoint presentation. (10)

   (b) Discuss the text formatting and alignment. (10)

7. (a) Explain the steps involved in creating and editing a query. (10)

   (b) Explain the method of creating and saving a table in Access. (10)

8. (a) How will you combine Excel data and charts with word documents? (10)

   (b) How will you display the records using a form? (10)

DE–481 **21**

DISTANCE EDUCATION

M.C.A. (S). DEGREE EXAMINATION, DECEMBER 2014

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) (i) Discuss the features of object–oriented programming. (5)

   (ii) How are data and functions organized in an object–oriented program? (5)

   (b) (i) Enumerate the rules of naming variables in C++. (5)

   (ii) List the tokens, keywords, identifier in C++. (5)

2. (a) Explain in detail about control structures in C++. (10)

   (b) Describe about Array and give an example for one dimensional and multidimensional array. (10)

3. (a) What is an expression? List the different types of expressions. Discuss any three with examples. (10)

   (b) (i) What do you mean by type compatibility? Explain with an example. (5)

   (ii) Write a short note on memory management operators. (5)

4. (a) What are the types of argument passing available in C++? Give example for all types. (10)

   (b) (i) How do structures differ in 'C' and 'C++'? (5)

   (ii) Explain overloaded function. (5)

5. (a) Write a C++ program for matrix multiplication. (10)

   (b) Describe the following with an example program.

   (i) Virtual function (5)

   (ii) Modularilation. (5)

6. (a) (i) Write a short note on constructors. (5)

   (ii) Write a macro to obtain the largest of three numbers. (5)

   (b) What are the merits and demerits of using friend function. (10)

7. (a) Discuss the following :

   (i) Polymorphism. (5)

   (ii) Function overloading. (5)

   (b) What is inheritance? Discuss all types of inheritance with example. (10)

8. (a) (i) Discuss about error handing during the file operation. (5)

   (ii) Describe the various classes for file operations. (5)

   (b) Write in detail about command line argument and illustrate with an example. (10)

DE–482 **22**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2014.

Second Semester

SYSTEMS ANALYSIS AND DESIGN

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain some characteristics that are present in all system. (10)

   (b) Explain the Multifaceted role of analyst. (10)

2. (a) Discuss top-down approach to system planning. (10)

   (b) Write in detail the steps in feasibility analysis. (10)

3. (a) Discuss about logical and physical design. (10)

   (b) Explain the various types of forms. (10)

4. (a) Explain sequential file organization. (10)

   (b) Discuss about warnier orr diagram. (10)

5. (a) Discuss the various training methods of a personnel. (10)

   (b) Explain corrective system maintenance. (10)

6. (a) Discuss about the elements of a system. (10)

   (b) Explain the various steps in system design. (10)

7. (a) Explain the formal and informal information systems. (10)

   (b) Write about the strategies for determining Information regarding user's requirement. (10)

8. (a) Explain the different steps that should be taken to a successful interview. (10)

   (b) What is a DFD? Explain different rules and methods to construct a DFD with example. (10)

DE–483 **23**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2014.

Second Semester

VISUAL PROGRAMMING – I

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain in detail about visual basic single document interface environment. (10)

   (b) Explain about the overview of how to develop a visual basic application. (10)

2. (a) Discuss in detail about toolbars and its use. (10)

   (b) Describe about custom controls and components in detail. (10)

3. (a) Explain in detail about textboxes and labels in VB. (10)

   (b) Write in detail about command button and option button in VB. (10)

4. (a) Explain VB operators with suitable example. (10)

   (b) Explain looping statements in visual basic with example. (10)

5. (a) Discuss about visual basic function in detail with example. (10)

   (b) Discuss about the data types in visual basic with example. (10)

6. (a) Explain in detail about user defined data type in visual basic. (10)

   (b) Write in detail about control arrays and dynamic arrays in visual basic with suitable example. (10)

7. (a) Discuss in detail about the types of objects in visual basic. (10)

   (b) Explain any ten visual basic graphics commands with example. (10)

8. (a) Explain Random Access File in visual basic with suitable example. (10)

   (b) Discuss about data control in visual basic with example. (10)

DE–484 **24**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2014.

Second Semester

Computer Applications

COMPUTER GRAPHICS

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

(5 × 20 = 100)

1. Write notes on:

   (a) Color CRT (12)

   (b) DDA algorithm. (8)

2. With program segment, explain the procedure of generating a line with Bresenham's line drawing technique. (20)

3. Illustrate and explain the concept of composite transformation techniques. (20)

4. Summarize the procedure of Cohen–Sutherland line clipping technique with program segment. (20)

5. With neat diagrams and matrices, explain the general three dimensional rotation transformation principle. (20)

6. Compare and Contrast parallel projections with perspective, projections that are applied to a three dimensional object. (20)

7. Write notes on:

   (a) Depth buffer method (12)

   (b) Depth sorting method. (8)

8. How do informations display and feedback mechanisms influence the user interface design? Explain in detail. (20)

DE–485 **31**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Third Semester

DISCRETE MATHEMATICS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) (i) Construct the truth table for the statement formula $P \vee \neg Q$.

(ii) Show that

$(\neg P \wedge (\neg Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R) \Leftrightarrow R$. (10)

(b) Show that the following are equivalent formulas

(i) $P \vee (P \wedge Q) \Leftrightarrow P$.

(ii) $P \vee (\neg P \wedge Q) \Leftrightarrow P \vee Q$. (10)

2. (a) Show that

$$(x)(P(x) \to Q(x)) \wedge (x)(Q(x) \to R(x)) \Rightarrow (x)(P(x) \to R(x))$$

(10)

(b) Show that $(\exists x)(P(x) \wedge Q(x)) \Rightarrow (\exists x)P(x) \wedge (\exists x)Q(x)$.

(10)

3. (a) (i) Show that $A \subseteq B \Rightarrow A \cap B = A$.

(ii) Show that for any two sets A and B $A - (A \cap B) = A - B$. (10)

(b) (i) Show that (1) $A - B = A \cap \sim B$ and (2) $A \subseteq B \Leftrightarrow \sim B \leq \sim A$.

(ii) If $A = \{\alpha, \beta\}$ and $B = \{1, 2, 3\}$. Find $A \times B$, $B \times A$, $A \times A$, $B \times B$ and $(A \times B) \cap (B \times A)$. (10)

4. (a) Let $R = \{<1,2>, <3,4>, <2,2>\}$.

$S = \{<4,2>, <2,5>, <3,1>, <1,3>\}$

Find $R \circ S$, $S \circ R$, $R \circ (S \circ R)$, $(R \circ S) \circ R$, $R \circ R$, $S \circ S$ and $R \circ R \circ R$. (10)

(b) (i) Define partially ordered set.

(ii) Let $X = \{2, 3, 6, 12, 24, 36\}$ and the relation $\leq$ be such that $x \leq y$ if x divides y. Draw the Hasse diagram of $< x, \leq >$. (10)

5. (a) If $f : x \to y$ is invertible then (i) $f^{-1} \circ f = I_x$ and (ii) $f \circ f^{-1} = I_y$. (10)

(b) Let $f : x \to y$ and $g : y \to z$ be both one to one and onto functions then $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. (10)

6. (a) Define characteristic function of a set.

(b) Write down its properties and prove $\psi_{A \cap B}(x) = \psi_A(x) * \psi_B(x)$.

(c) Prove that $\sim\sim A = A$. (20)

7. (a) (i) Define submonoid.

(ii) Prove that for any commutative monoid $< M, * >$ the set of idempotent elements of M forms a submonoid. (10)

(b) (i) Define subgroup of $< G, * >$.

(ii) Prove that A subset $S \neq \phi$ of G is a subgroup of $< G, * >$ iff for any pair of elements $a, b \in S$, $a * b^{-1} \in S$. (10)

8. (a) Show that the sum of Indegrees of all the nodes of a simple digraph is equal to the sum of out degrees of all its nodes and this is equal to the number of edges of the graph (10)

(b) Define (i) Directed tree (ii) Ordered tree (iii) Binary tree (iv) Adjacency matrix (v) Path matrix. (10)

DE–486 **32**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A (S)/(Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Third Semester

OPERATING SYSTEMS

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Briefly explain the history of operating systems. Discuss different types of operating systems and compare them. (10)

   (b) Describe the functions and services of an operating system. (10)

2. (a) Explain about Priority scheduling and round robin scheduling. Compare their performance. (10)

   (b) Explain about Semaphores (10)

3. (a) Write short notes about Race conditions (10)

   (b) Discuss about message passing. (10)

4. (a) What is Deadlock? Explain the principles and conditions for deadlock. (10)

   (b) Describe the functions of the Device independent I/O software. (10)

5. (a) Explain the difference between internal and external fragmentation. (10)

   (b) Compare paging with segmentation (10)

6. (a) Explain any two page replacement algorithms. (10)

   (b) Discuss the design issues for paging systems. (10)

7. (a) What is a file? Explain the different access methods for files. (10)

   (b) Write short notes on Directory structure. (10)

8. (a) Explain about Disk space management. (10)

   (b) Explain about various protection mechanisms in File management. (10)

DE–487 **33**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Third Semester

RDBMS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Define Database. Discuss the features of DBMS? (2 + 8)

   (b) Write short notes about different user interfaces.(10)

2. (a) Discuss the advantages and disadvantages of DBMS. (10)

   (b) Describe the important Components of DBMS. (10)

3. (a) Explain about data definition facilities in relational model. (10)

   (b) Discuss about the concepts of relational model. (10)

4. (a) Write short notes about mapping operations in relational model. (10)

   (b) Explain about data manipulation facilities with an example. (10)

5. (a) Explain the types of databases. (10)

   (b) Write short notes about ORACLE database architecture. (10)

6. (a) Briefly explain about the elements of SQL language. (10)

   (b) Explain about DML commands. (10)

7. (a) Discuss about SET operators in SQL. (10)

   (b) List some basic queries. Assume your own data. (10)

8. (a) Explain about Table citation with an example. (10)

   (b) Discuss about the following with an example

      (i) Modifying a table (5)

      (ii) Copying and renaming a table. (5)

DE–488 **34**

DISTANCE EDUCATION

M.C.A. (S) / M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Third Semester

UNIX AND SHELL PROGRAMMING

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What are the Common file system commands? Explain.

   (b) Explain about file permissions.

2. (a) Explain the following filters:

   (i) grep

   (ii) egrep

   (iii) fgrep.

   (b) Distinguish between ***ed*** and ***awk*** utilities available in Unix.

3. (a) List and explain the shell metacharacter.

   (b) Write note on shell variables with example.

4. (a) What is the use of ***cal*** command? Explain the customizing the ***cal*** command.

   (b) Illustrate the ***while*** and ***until*** loops with example.

5. (a) Explain the following command with example.

   (i) pick

   (ii) news

   (b) What are the commands used for tracking file changes? Explain.

6. (a) Explain the file access through shell program with example.

   (b) Explain the role of following commands in UNIX environment.

   (i) fork

   (ii) wait

7. (a) What is ***yacc***? Write the overview of ***yacc***.

   (b) Explain about ***ms*** macro package.

8. Write a menu driven shell program for the following.

   (a) List of files

   (b) Processes of users

   (c) Current date

   (d) Users of system

   (e) Quit to Unix.

DE–489 **35**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Third Semester

COMMUNICATION SKILLS

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the levels of communication in a detailed manner. (10)

   (b) What are the barriers to communication and explain the classification of barriers? (10)

2. (a) What is mean by reading comprehension and the reasons for poor comprehension and how can it be improved? (10)

   (b) Explain the effective writing, report writing and speech writing. (10)

3. (a) What is mean by cover letter and why is it important and explain it with a example? (10)

   (b) How can we develop the ability to listen patiently and what are the steps to be followed for effective listening? (10)

4. (a) Discuss the methods of asking questions using telephone and about the different kinds of calls. (10)

   (b) What are the guidelines for planning and conducting a interview and also the steps to be followed for a successful interview? (10)

5. (a) Why does group discussion depends on non-verbal skill and explain its importance? (10)

   (b) What is negotiation and explain the critical features of negotiation? (10)

6. (a) What are the types of questions can be asked during the course of an interview and what are the answering techniques? (10)

   (b) Explain about the group discussion and the methods to be followed for a successful group discussion. (10)

7. (a) Write about the structure and organisation of a report. (10)

   (b) Explain the mock meetings and seminars. (10)

8. (a) Discuss the communication within management. (10)

   (b) Describe the various techniques for delivering presentation and model presentation. (10)

DE–490 **41**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fourth Semester

COMPUTER NETWORKS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Describe the TCP/IP Reference model. (12)

   (b) Discuss about the telephone system. (8)

2. (a) Explain the architecture of ATM. (10)

   (b) Discuss on Error detection and correction. (10)

3. (a) Bring out the concept of collision free protocols. (10)

   (b) Explain about the media access protocols. (10)

4. (a) Discuss about the routing algorithms of Network layer. (10)

   (b) Explain about the Internet Control Protocol. (10)

5. (a) Explain about Routing and Switching. (10)

   (b) Discuss in detail about crash recovery. (10)

6. (a) Describe about Internet Transport Protocols. (10)

   (b) Write the protocols for Gigabit Networks. (10)

7. (a) Give the details about Network Security. (10)

   (b) Discuss about DNS. (10)

8. (a) Explain Substitution and Transposition ciphers. (10)

   (b) Write a note on Video data compression. (10)

DE–491 **42**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fourth Semester

ACCOUNTING AND FINANCIAL MANAGEMENT

(2005–2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. XY Ltd has purchased and issued the material 'M' in the following order :

December	1	Purchases 300 units at Rs. 3
	4	Purchases 600 units Rs. 4
	6	Issue 400 units
	10	Purchase 600 units at Rs. 4
	15	Issue 1,000 units
	20	Purchase 400 units at Rs. 5
	23	Issue 200 units

Prepare a stores ledger by adopting FIFO, LIFO METHOD of issuing materials.

2. The following transactions, Prepare Journal, ledger and Trial Balance

2005 April	1st	Arul commenced business with Rs. 21,000
	2nd	Bought goods for cash Rs. 9,200
	5th	Purchase goods from Mani Rs. 3,300
	7th	Sold good to Kannan on credit Rs. 5,600
	9th	Received cash from Kannan Rs. 3,600
	11th	Paid Mani on account Rs. 2,100
	16th	Sold goods to Manokar Rs. 3,500
	21st	Cash sales Rs. 7,500
	26th	Paid Mani Rs. 1,200
	29th	Received cash from Manokar Rs. 3,500
	30th	Paid rent Rs. 450
	30th	Paid salary to office staff Rs. 700

3. The balance sheet of AB and Co. Ltd .., at the end of 1968 and 1969 are given below.

Liabilities	1968 (Rs.)	1969 (Rs.)	Assets	1968 (Rs.)	1969 (Rs.)
Share capital	1,00,000	1,50,000	Freehold land	1,00,000	1,00,000
Share premium	–	5,000	Plant at cost	1,04,000	1,00,000
General reserve	50,000	60,000	Furniture at cost	7,000	9,000
Profit and loss A/C	10,000	17,000	Investment at cost	60,000	80,000
6% Debentures	70,000	50,000	Debtors	30,000	70,000

Liabilities	1968 (Rs.)	1969 (Rs.)	Assets	1968 (Rs.)	1969 (Rs.)
Provision for depreciation on plant	50,000	56,000	Stock	60,000	65,000
Provision for depreciation on furniture	5,000	6,000	Cash	30,000	45,000
Provision for taxation	20,000	30,000			
Sundry creditors	86,000	95,000			
	3,91,000	4,69,000		3,91,000	4,69,000

Additional Information ;

(a) A plant purchased for Rs. 4,000 (depreciation Rs. 2,000) was sold for cash Rs. 800 on 30.9.1969

(b) On 30th June 1969, an item furniture was purchased for Rs. 2,000. These were the only transactions concerning fixed assets during 1969

(c) Depreciation on plant was provide at 8% on cost (the sold out item is not taken into consideration) and on furniture at 12 .5 % on average cost

(d) A dividend of 22.5 % on original shares was paid.

You are required to prepare funds flow statement.

4. The following is the trial balance sheet as on 31st December 2011.

Debit balance	Rs.	Credit balance	Rs.
Opening inventory	72,000	Capital	5,00,000
Purchases	2,25,000	Sales	3,50,000
Furniture	15,000	Purchase return	1,800
Motor car	30,000	Ram	32,000
Buildings	4,25,800	Commission	31,500
Sanjay	12,000		
Kishore	20,000		
Raghav	18,000		

Debit balance	Rs.	Credit balance	Rs.
Advertisement	22,000		
Repairs and maintenance	13,000		
General expenses	16,000		
Insurance	7,000		
Cash in hand	3,500		
Cash at bank	6,000		
Salaries	30,000		
Total	9,15,300	Total	9,15,300

You are required to draft profit and loss account and balance sheet as on 31st December 2011 after taking into account the following adjustments :

(a) Closing inventory as at 31.12.2011

(b) Interest on capital at 6%

(c) Prepaid advertisement Rs. 2,000

(d) Goods used for domestic purpose Rs. 1,800

(e) Outstanding salaries Rs. 3,000

(f) Depreciation on buildings at 5% on furniture at 5% and motorcar at 10%.

5. From the following information relating to Standard Engineering Co., Prepare a cash budget for the half year ending 30th September, 1998

Month	Sales	Material	Wages	Overheads
March	1,50,000	90,000	18,000	31,000
April	1,40,000	1,20,000	28,000	35,400
May	1,54,000	84000	20800	36,500
June	1,68,000	1,36,000	22,200	36,500
July	1,82,000	1,28,000	30,000	37,500
August	2,04,000	1,51,000	28,000	38,000
September	2,40,000	1,62,000	25,000	38,000

Other information :

(a) Bank balance on 1.4.98 was Rs. 72,000

(b) The loan installment of Rs. 25,000 per month was to be remitted to the distributors

(c) Sales commission @ 5% on sales was to be paid in the month following the sales

(d) Rs.50,000 was expected to be collected from the customers as deposit in July

(e) 50% of the sales are on credit and the period of credit being one month

(f) 50% of the material can be paid for after a month

(g) 50% of the overheads could be paid in the month following their incidence

(h) Wages are paid on the first working day of the following month.

6. Define "Overhead". Explain the different methods of classifying overheads with suitable examples.

7. Give the purpose and method of computation of the following :

(a) Return on Equity

(b) Operating profit ratio

(c) Fixed assets turnover ratio

(d) Average Collection period

(e) Creditors turnover ratio

(f) Capital gearing ratio.

8. The sales and profit during two years were as follows :

Year	Sales (Rs.)	Profit (Rs.)
1988	1,50,000	20,000
1989	1,70,000	25,000

You are required to calculate :

(a) The P/V ratio

(b) The Breakeven point

(c) The sales required to earn a profit of Rs. 40,000

(d) The profit made when sales are Rs. 40,000

(e) The Margin of safety at profit of Rs. 50,000 and

(f) Variable cost of two years.

DE–493 **44**

DISTANCE EDUCATION

M.C.A. (S)/ M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fourth Semester

OBJECT ORIENTED ANALYSIS AND DESIGN

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the elements of an object model. (10)

   (b) How does object oriented approach different from traditional approach in system development? (10)

2. Describe the nature of an object. Explain the relationships among objects with an example. (20)

3. What is classification? Explain the importance of proper classification. (20)

4. (a) How to identify proper classes and objects. Explain. (10)

   (b) Write short note about key abstraction mechanism. (10)

5. (a) Explain the concept of Unified Modeling Language (UML). (10)

   (b) Explain the purpose of Analysis and Design. (10)

6. (a) Describe the process of Inception and Elaboration. (10)

   (b) Write short note about Patterns and Transmission. (10)

7. (a) Briefly explain the essentials of class diagram. (10)

   (b) Draw class diagram for Student Information System. (10)

8. Explain the uses of Interaction diagram. Draw Interaction diagram for Library Management. (20)

DE–494 **45**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fourth Semester

Computer Applications

INTERNET PROGRAMMING

(2005 to 2010 Calender year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain developing intranet applications. (10)

   (b) How to write CGI script in perl? (10)

2. (a) Discuss about the objects in Javascript. (10)

   (b) Explain the control statements in Javascript. (10)

3. (a) Explain selection statements in VB script. (10)

   (b) Describe events in VB script. (10)

4. (a) Explain HTML basic tags. (10)

   (b) Write a note on SGML. (10)

5. (a) Explain ActiveX control. (10)

   (b) How to create plugins in Netscape navigator? (10)

6. (a) Discuss about table tags in HTML. (10)

   (b) Write a note on Shockwave. (10)

7. (a) Describe the key elements of web page design. (10)

   (b) Explain hyperlinks and images. (10)

8. (a) What are the guidelines of graphics? Explain. (10)

   (b) Explain how can you add graphics to web pages. (10)

DE–495 **51**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fifth Semester

IMAGE PROCESSING AND ANALYSIS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the elements of digital geomentry in detail. (10)

   (b) Explain about the motivation and perspectives of Image processing. (10)

2. (a) Discuss about the texture and pattern discrimination in image processing system. (10)

   (b) Explain the application of digital image processing. (10)

3. (a) Discuss about the contrast intensification and smoothing. (10)

   (b) Discuss about the low-pass and high –pass filtering. (10)

4. (a) Write in detail about Homorphic filtering. (10)

   (b) Describe Restoration by maximum posterior estimation. (10)

5. (a) Explain about loss- less compression. (10)

   (b) Write in detail about Run-length coding. (10)

6. (a) Explain in detail about plane – to – plane transformation mapping. (10)

   (b) Write in detail about the processing of colour images. (10)

7. (a) Write in detail about any one of the segmentation method in the image processing. (10)

   (b) Describe in detail about the threshold selection methods. (10)

8. (a) Discuss about the pattern filtering approach. (10)

   (b) Describe the elements of extraction by thresholding. (10)

DE–496 **52**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fifth Semester

SOFTWARE ENGINEERING

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Define Software Engineering and explain about the development of process model. (10)

   (b) Explain about the software metrics in detail with examples. (10)

2. (a) Explain the size factors, quality and productive factors in detail. (10)

   (b) Discuss about role of management in software development. (10)

3. (a) Discuss about estimating software scope, resources and project estimation. (10)

   (b) Describe the programming team structure in detail. (10)

4. (a) Discuss about software reviews. (10)

   (b) Describe the planning for quality assurance. (10)

5. (a) Explain the software configuration management. (10)

   (b) Describe the risk mitigation, monitoring and management in detail. (10)

6. (a) Explain the information flow in detail with diagram. (10)

   (b) Discuss about the structured analysis. (10)

7. (a) Describe in detail about structure specification techniques. (10)

   (b) Explain about Architectural Design. (10)

8. (a) Discuss about the software testing method. (10)

   (b) Describe the unit testing and Integration testing in detail. (10)

———————

DE–497 **53**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fifth Semester

VISUAL PROGRAMMING — II

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss in detail about the window message. (10)

   (b) Explain in detail about the file creation using Appwizard. (10)

2. (a) Write in detail about mainframe object and view object. (10)

   (b) Discuss in detail about VC++ views and documents. (10)

3. (a) Explain in detail about adding accelerator keys. (10)

   (b) Write in detail about how menu items connected to code with example. (10)

4. (a) Describe in detail about menus. (10)

   (b) Explain in detail about mouse events. (10)

5. (a) Explain in detail about dialog boxes. (10)

   (b) Write a VC++ program to create simple calculator. (10)

6. (a) Write in detail about combo boxes and list boxes. (10)

   (b) Discuss about VBX control in detail. (10)

7. (a) Explain in detail about Random Access file with example. (10)

   (b) Write short note on the following :

      (i) adding scroll bars (5)

      (ii) splitter windows. (5)

8. (a) Explain the concept of exception handling in detail with suitable example. (10)

   (b) Write a short note on following :

      (i) Creating Employee database. (5)

      (ii) Adding and Deleting records. (5)

DE–498 **54**

DISTANCE EDUCATION

M.C.A. (S) /M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fifth Semester

COMPILER DESIGN

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. Explain in detail about :

   (a) Lexical analysis (10)

   (b) Syntax analysis. (10)

2. (a) Write a detailed note on top-down parsing. (10)

   (b) Explain the implementation of LR parsing tables. (10)

3. (a) Explain with suitable examples the construction of parse tree. (15)

   (b) Write short notes on automatic parser generator. (5)

4. (a) Explain in detail about implementation of syntax-directed translators. (10)

   (b) Write short notes on three-address code and quadruples. (10)

5. Describe in detail about the contents and data structures of a symbol table. (20)

6. (a) Explain briefly about lexical phase errors. (10)

   (b) Explain briefly about semantic errors. (10)

7. (a) Write short notes on Loop optimization. (10)

   (b) Explain in detail about a simple code generator. (10)

8. (a) Explain the role of lexical analyzer. (10)

   (b) Write short notes on regular expressions. (10)

———————

DE–499 **55**

DISTANCE EDUCATION

M.C.A.(S)/M. C.A.(S) (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fifth Semester

DATA WAREHOUSING AND MINING

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Write notes on self learning computer systems. (10)

   (b) Discuss on Data mining vs. query tools. (10)

2. (a) Discuss on data enrichment and coding. (10)

   (b) Explain visualization techniques in detail. (10)

3. (a) Write in detail about Decision trees. (10)

   (b) Discuss on OLAP tools. (10)

4. (a) With a neat sketch, explain 3-tier Data Warehouse architecture. (10)

   (b) Discuss on system process and process architecture. (10)

5. (a) Describe Meta Data and its uses. (10)

(b) Write notes on systems and data warehouse process managers. (10)

6. (a) Explain about hardware architecture of Data Warehouse. (10)

(b) Discuss on security features in Data Warehouse. (10)

7. (a) Explain the backup strategies of data warehouse. (10)

(b) Write notes on operating the data warehouse. (10)

8. (a) Discuss on capacity planning. (10)

(b) Write notes on testing the Data Warehouse. (10)

DE–500 **61**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Sixth Semester

SOFTWARE PROJECT MANAGEMENT

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. Explain the project plan in detail with suitable diagrams. (20)

2. Explain project analysis and technical planning in detail. (20)

3. What is sequencing and scheduling projects? Explain with examples. (20)

4. Explain how to organic people and teams in projects with suitable illustrations. (20)

5. What are the various standards in software configuration? (20)

6. Explain the PRINCE project management with necessary illustrations. (20)

7. What is prototyping? Explain the various models with suitable examples. (20)

8. Explain resource allocation with a neat diagram. (20)

DE–501 **62**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Sixth Semester

MULTIMEDIA SYSTEMS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the benefits and problems of multimedia in detail. (10)

   (b) Explain the system components in detail. (10)

2. (a) Explain about the types of development tools in multimedia in detail. (10)

   (b) Discuss about multimedia platforms. (10)

3. (a) Discuss about digital ink and speech audio. (10)

   (b) Compare analog versus digital video and explain the steps to add a sound to the multimedia project. (10)

4. (a) Discuss about MPEG and JPEG. (10)

   (b) Briefly explain about the different techniques used for video compression. (10)

5. (a) Explain about DVI standards. (10)

   (b) Explain about redundancy and visibility in detail. (10)

6. (a) Explain the object and classes in object oriented multimedia frameworks in detail. (10)

   (b) Discuss about format classes in multimedia frameworks. (10)

7. (a) Explain about media classes in multimedia frameworks. (10)

   (b) Explain about transform classes in multimedia frameworks. (10)

8. (a) Explain the architecture and operations in multimedia environments in detail. (10)

   (b) Explain how the multimedia can be used in training and education. (10)

DE–502 **63**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Sixth Semester

NEURAL NETWORKS

(2005 to 2010 calender year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. Discuss about the following :
   - (a) Activation function (5)
   - (b) Single layer feed forward network (5)
   - (c) Characteristics of neural networks. (10)

2. (a) Write in detail about the learning tasks. (10)
   - (b) Write about error-correction learning. (10)

3. (a) Discuss about Hebbian learning methodology. (10)
   - (b) Explain the competitive learning methodology. (10)

4. (a) Discuss about single layer perceptrons. (10)
   - (b) Discuss about the applications of counter propagation networks. (10)

5. (a) Explain the structure of the Hopfield networks. (10)
   - (b) Explain the concept of associative memory. (10)

6. (a) Explain how neural networks are constructed. (10)

   (b) What are the applications of artificial neural networks? (10)

7. (a) Give an overview on adaptive resonance theory. (10)

   (b) Discuss about optical neural networks. (10)

8. (a) Explain holographic correlators. (10)

   (b) Discuss about Neocognitrons. (10)

DE–503 **64**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Sixth Semester

C – SHARP (C#)

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Write in detail about common language runtime. (10)

   (b) Discuss about the evolution of C#. (10)

2. (a) Explain in detail about error handling and deployment in C#. (10)

   (b) Discuss in detail about code reuse and security in C#. (10)

3. (a) Explain in detail about C# features. (10)

   (b) Explain about the virtual object system in C#. (10)

4. (a) Explain in detail about structure in C# with example. (10)

   (b) Explain about the boxing and unboxing in C# in detail. (10)

5. (a) What do you mean by Constructor? Explain constructor in C# with example. (10)

   (b) Explain method overloading in C# with suitable example. (10)

6. (a) Explain in detail about exception handling in C# with suitable example. (10)

   (b) Discuss in detail about iteration statements in C# with example. (10)

7. (a) Describe about C# components building and compiling. (10)

   (b) Explain the creation of simple client in C# with example. (10)

8. (a) Write in detail about platform invocation services in C#. (10)

   (b) Write short note on following in C# :

      (i) Intermediate Languages. (5)

      (ii) Disassembler. (5)

DE–8836 **15**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2014.

First Semester

Computer Application

Lab I — DATA STRUCTURES USING 'C'

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Examiner should select and give one question to each candidate by lot system.

Break-up of marks :

Record Note Book	: 5
Algorithm/Flow chart	: 15
Program	: 35
Debugging and Execution	: 35
Result	: 10
Total	: 100

1. (a) Write a menu program in C to find an element using Linear and Binary search methods.

   (b) Write a C program to implement all string operations using switch statement.

---------- Cut here ----------

2. (a) Write a C program to copy contents of one file to another file.

   (b) Write a C program to sort a set of elements using insertion sort.

3. (a) Write a C program to implement insert and delete operations on queue using array concept.

   (b) Write a C program to find the roots of a Quadratic equation $aX^2 + bX + c = 0$.

---------- Cut here ----------

4. (a) Write a C program to implement push and pop operations on stack.

   (b) Write a C program to implement all string operations using switch statement.

---------- Cut here ----------

5. (a) Write a C program to find the factorial of a given number using recursion.

   (b) Write a C program to find the roots of a Quadratic equation using function definition $aX^2 + bX + c = 0$.

---------- Cut here ----------

6. (a) Write a C program to evaluate the given mathematical expression using stack.

   (b) Write a C program to sort 10 Nos. in ascending order with naming of variable and the value before and after sorting.

---------- Cut here ----------

7. (a) Write a C program to read 10 values to an array variable. Use pointers to locate and display each value.

   (b) Write a C program for linked list implementation of Queue operations.

8. (a) Write a C program to implement push and pop operations on stack.

(b) Write a C program to convert a given number into words for number 1 to 5.

(Eg : 1 to ONE, 2 to TWO and 5 to FIVE)

DE–8837 **16**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2014.

First Semester

Computer Application

LAB II — MS OFFICE AND INTERNET

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any TWO questions, choosing ONE from Section A and another from Section B.

Each question carries 50 marks.

Break-up marks

Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

SECTION A

1. Type a lecture note of your own and provide audio explanation with the help of sound files.

---------- Cut here ----------

2. Prepare your resume with your photograph inserted use table, bullets and different color features.

3. Create two tables SALES 1, SALES 2 and join them to produce a third table SALES 3.

---------- Cut here ----------

4. Prepare a news report using two columns insert a picture in the first column and make the text flow around it.

---------- Cut here ----------

5. Prepare an invitation for a function to be conducted in your institution use different text orientation and pictures to make it attractive.

---------- Cut here ----------

6. Draw LINE, BAR, PIE charts for the data given below :

ABC Company Limited

Year	Sales	Cost	Profit
1991	1000	400	600

Add data for 10 years from 1991 to 2000 provide titles, legends, grids and data labels.

---------- Cut here ----------

7. Create a form letter that informs the customer about the date of maturity of a deposit amount in a commercial bank and request the customer for renewal mail merge it with an Access Database containing all the customer data. Prepare letters for customers whose due date falls in a specific range of dates.

8. Create student table with the following fields REGNO, NAME, MARK 1, MARK 2, MARK 3. Create ADDRESS table with fields REGNO, STREET, CITY and PIN.

   (a) Write a query to display REGNO, NAME and total of all the three subject marks.

   (b) Write a query to display REGNO, NAME, STREET, CITY and PIN and total of all the three subject marks.

---------- Cut here ----------

SECTION B

9. Create HTML page that plays video and audio files.

---------- Cut here ----------

10. Create HTML page to display the pictures of Elephant and Monkey. Link to another page when user clicks over the picture the linked page should display features of the animal clicked.

---------- Cut here ----------

11. Create HTML page to display 10 courses and their highlighted features, conducted by an educational institution in a colorful format.

DE–8838 **25**

DISTANCE EDUCATION

M.C.A. DEGREE EXAMINATION, DECEMBER 2014.

Second Semester

Practical

COMPUTER LAB III — GRAPHICS USING C++

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Examiner has to choose the ONE question for each candidate by lot system.

Break-up marks

Record Note Book	5
Algorithm/Flow chart	15
Program	35
Debugging and Execution	35
Result	10
Total	100

1. Create a base class shape, which contains the common elements of circle, triangle and rectangle, from which you derive three classes called ball, rect and triangle. Include two member functions set( ) to set coordinates and draw( ) which draws them. Write a main( ) program to exercise this class using graphic-mode graphic functions.

2. Develop a program to do the following :

   (a) Draw a triangle, translate it to a distance and then rotate it through an angle clockwise.

   (b) With the same triangle, reverse the operations in case (a)

   (c) Check whether the result figures of case (a) and case (b) are one and the same or different.

---------- Cut here ----------

3. Write a C++ program to show graphical transformation (scaling, rotation translation) of a two dimensional object.

---------- Cut here ----------

4. Write a C++ program to show graphical transformation on three dimensional object.

---------- Cut here ----------

5. Implement the polygon clipping algorithm using C++ graphics mode graphics functions.

---------- Cut here ----------

6. Implement the Bresenham's algorithm for drawing circle and ellipse using C++ in graphics mode.

---------- Cut here ----------

7. Write a C++ program to draw non-linear objects arcs and curve.

---------- Cut here ----------

8. Draw a ball perform animation on it.

9. Draw any 3-D object and perform animation on it.

---------- Cut here ----------

10. Write procedure and a C++ program to draw nonlinear object ellipse.

DE–8839 **26**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, DECEMBER 2014.

Second Semester

Computer Application

LAB IV — VISUAL BASIC

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Each subdivision carries 50 marks.

Break-up of marks
Record Note Book : 5
Algorithm/Flow chart : 15
Program : 35
Debugging and Execution : 35
Result : 10
Total : 100

1. (a) Design and implement one Active X control for monthly Calendar.

   (b) Write a VB project to display a message (Example Left/Right mouse button is clicked) When the mouse button is clicked using mouse event of the form.

2. (a) Create an application using VB to move elements from one list to another list and add new items.

   (b) Create a VB application to make the text bold, Italic, underline and also to color the text (using check box and option button).

---------- Cut here ----------

3. (a) Design and implement one Active X control for stop watch used for sports.

   (b) Design and implement and Active X control for Railway Clock (Hour display 1 to 24)

---------- Cut here ----------

4. (a) Write a VB project using in Active X control (Rich text box), develop the windows NOTEPAD like editor with file and edit menus and also display floating menu whenever necessary.

   (b) Design and implement one Active X control for digital electric current usage measurement meter used in house.

---------- Cut here ----------

5. (a) Write an VB project to calculate the amount when the user selects the class type of the train from the combo box and also enters the number of passengers.

   (b) Write a VB project for a blood bank that maintains a list of donors with address and their blood group provide the following reporting features. Display all donors (using data reports).

      (i) In age group 20-30

      (ii) In a particular city.

6. (a) Write a VB project to create a screen saver that displays a list of pictures with 1 second pause in between successive pictures.

   (b) Write a VB project that receives a year number from a text box and displays number of days in the given month. Take care of leap years. Use lost focus event for list box.

---------- Cut here ----------

7. (a) Write a VB project using control array that creates an ordinary calculator with appropriate command buttons. Include the following capabilities for the calculator. +, -, *, 1, -1.

   (b) Write a VB project that stores 10 employee records with field EMPNO, NAME, AGE, SEX and SALARY. In an array display data fields in text boxes and provide command buttons to move to desired record.

---------- Cut here ----------

8. (a) Write a VB project for commercial bank operations using SB account database with the following features.

      (i) Deposit amount

      (ii) With draw amount (with minimum balance condition)

   (b) Write a VB project to display the availability of seats when the user select the Bus name and number from the combo box and timings from another combo box.

9. (a) Write a VB project to display the availability of books when the user select the author name from the combo box and book title from another combo box.

(b) Design and implement one Active X control for digital electric current usage measurement meter used in houses.

---------- Cut here ----------

10. (a) Write a VB project to display the availability of seats when the user selects the flight name from the combo box and date including timings from another combo box.

(b) Write a VB project using AX DLL or EXE add a class module that would perform the following functions.

(i) Sum of digits

(ii) Whether the given number is Armstrong or not.

DE–8840 **36**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Third Semester

LAB V — RDBMS

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Students has to answer ONE question by lot system.

Break up of marks :

Record Note Book	: 5
Algorithm/Flow chart	: 15
Program	: 35
Debugging and Execution	: 35
Result	: 10
Total	: 100

1. A hospital maintains blood donors record information in a file. The file contains the following fields :

   Donor number, donor name, donor age, donor address, pin code, place of birth, blood group(A. B, AB & O).

   Write a program using Oracle to print out the number, name and address of the donors for the following categories :

   (a) Blood donors having AB blood group

   (b) Blood donors in the age group between 16 to 25

   (c) Female donors with 'A' blood group in the age between 18 and 25.

2. Write a PLSQL program to prepare a Mark statement of a student using student table and Mark table. Assume your own data.

---------- Cut here ----------

3. Write a program in SQL for hospital Billing system with the following fields :

Patient no, patient name, age, Attending doctor. patient–type (IN/OUT), consulting charge, Blood test charge, X-Ray charges, other test charges and total fee.

Prepare reports with the following conditions :

(a) The patients undergone blood test

(b) The patients who taken X-rays

(c) The patients who belong to In-patient category.

The common fields to be included in the above mentioned reports :

Patient No., Name, Age, Charges and Total Charges.

4. A News paper vendor buys newspapers on wholesale from a distributor for Rs. 4 and sell them in retail for Rs. 4.50. At the end of the day the unsold papers are returned to the distributor for a Rs. 1 rebate per paper. Write a program using Oracle to prepare a port for the newspaper vendor in the following format with 9 weeks data :

Week	Bought	Sold	Return	Profit	Loss
1	100	50	XX	XX.XX	XX.XX
2	75	60	XX	XX.XX	XX.XX
--	--	---	---	-----	-----
--	--	---	---	-----	-----
Total					

5. A salary statement contains Name, Basic pay, Allowances, Total Deduction (includes IT), Gross pay, Net pay.

Dearness Allowance = 58% of Basic pay

Gross Pay = Basic pay + D.A

Deduction = P.F + LOAN AMT

Income Tax is calculated on the basis of annual income with the following conditions.

Annual salary	Income Tax
≤ 1 Lakh	Nil
> 1 Lakh but ≤ 1.5 Lakh	10%
> 1.5 Lakh but ≤ 3 Lakh	20%
> 3 lakh and above	30%

Write a program in SQL to prepare Salary report for 25 employees.

---------- Cut here ----------

6. A daily sales file contains records with the following fields :

Department number, date, item description, sale price for each item, quantity, cost of each item. Write a program using Oracle to list all the input data and compute total amount of sales and profit. Show the output report in the following format :

Department number

Item description

Sales price

Quantity

Cost

Profit

7. Write an SQL program for the canteen Maintenance with two tables namely MENU and BILL.

The MENU table contains the following fields :

ITEM, UNIT-Rate.

The following items are available in the canteen :

TEA, COFFEE and COOL-DRINKS

The BILL table contains the following fields :

EMPNO, NAME, DATE OF ISSUE, ITEM1, NO.OF TOKENS ISSUED FOR ITEM1, RATE1, ITEM2, NO.OF TOKENS ISSUED FOR ITEM2, RATE2, ITEM3, NO.OF TOKENS ISSUED FOR ITEM3, RATE3, TOTAL AMOUNT.

Note : An employee is allowed to get the token only once in a month.

RATE = UNIT – RATE * NO.OF TOKENS FOR PARTICULAR ITEM

TOTAL AMOUNT = RATE-1 + RATE-2 + RATF-3

Prepare a report in the following format :

S.NO.	EMP.NO	EMPNAME	PURCHASED RATE
1	--	--	--
2	--	--	--
3	--	--	--
			Total Amount :

8. An airline reservation database contains the reservation table and personal table.

Reservation table contains following fields :

Flight Number, Class, Passenger Name, Seat Number

Personal Table contains following fields :

Passenger Name, Sex, Age, Marital status, Nationality

Write a program in ORACLE to prepare the following reports :

(a) List the passengers' names with seat number according to Class and flight numbers.

(b) Total number of married female passengers in a particular flight.

(c) List out the Male passengers in the age between 18 to 25 for all flights.

---------- Cut here ----------

9. A company RX Pvt. Ltd states monthly salary to its employee. It consists of Basic salary, Allowance and deduction.

DA. – 43% of Basic Salary

H.R.A. – 7% of (Basic Salary + D.A)

Deductions :

P.F. – subscribed by an capital

LIC premium – payable by an employee, the salary saving scheme.

Loan premium – If any payable by an employee

Create a master of 10 records with following fields :

Employee number, name, designation, Basic Salary, HRA, DA, Bank A/c number, LIC premium number.

A transaction tile contains :

Employee number, P.F. subscription, LIC premium amount, Loan recovery.

Write a program in SQL to prepare a report with the following information :

Serial number, Employee number, Bank A/c number, Name, Basic Salary, Total Allowances, Gross pay, Total deductions, Net pay.

10. An examination has been conducted to a class of 55 students and the following are given as inputs :

Three test marks, Register No. and Candidate Name.

Write a SQL program to do the following :

Assign a letter grade to each student based on the average score and list Register Number, Average score, Grade. The minimum pass for each subject is 50.

The grading System :

Average score	Grade
90-100	A
75-89	B
60-74	C
50-59	D
0-49	F-Fail

DE–8841 **37**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION,
DECEMBER 2014.

Third Semester

Computer Applications

LAB VI — ALGORITHM AND SHELL PROGRAMMING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

ONE question is to be given by the examiner to each candidate by lot system.

Break-up marks

Record Note Book	: 5
Algorithm/Flow chart	: 15
Program	: 35
Debugging and Execution	: 35
Result	: 10
Total	: 100

1. (a) Write a C++ program to implement the Towers of Hanoi of problem.

   (b) Write a Shell program for 4-functions calculator using yacc command.

2. (a) Write a C++ recursive program to perform the following :

   (i) Factorial

   (ii) Fibonacci series.

   (b) Write a shell program to send mail to groups of users by extracting their IDs from/etc/group file.

---------- Cut here ----------

3. (a) Write a C++ program to search a number using Linear and Binary search method.

   (b) Write a shell program to do the following :

   (i) Display the string in center with BOLD and NORMAL.

   (ii) Display the string at given ROW, COLUMN with BOLD and NORMAL.

   (iii) Search a given string using awk and various options.

---------- Cut here ----------

4. (a) Write a C++ program to implement the Travelling Salesman problem using Branch and Bound method.

   (b) Write a Shell program to sum up the series.

$$\frac{1}{1!}+\frac{1}{2}+\frac{1}{3!}+....+\frac{1}{10!}$$

5. (a) Write a C++ program to implement the Knight Tour problem using backtracking method.

(b) Write a shell program to accept the name of a file from standard input media and then perform the following tests on it.

(i) Enter the 5 names in a file

(ii) Sort the names in existing file

(iii) List sorted and unsorted file

(iv) Quit.

---------- Cut here ----------

6. (a) Write a C++ program sort a list of numbers in ascending order using Heap sort method.

(b) Write a menu-driven shell program to perform the following file operations :

COPY, EDIT, RENAME, and DELETE.

---------- Cut here ----------

7. (a) Write a C++ program to sort a list of numbers in ascending order using quick sort (Divide and Conquer) method.

(b) Write a menu-driven shell program to perform the following tasks :

(i) Enter the sentence in a file

(ii) Search a word

(iii) Quit.

8. (a) Write a C++ program to sort a list of numbers in ascending order using straight exchange sorting method.

(b) Write a Shell program to perform the following tasks in Hospital maintenance system.

(i) List patents in Blood group wise

(ii) List patients in age between 20 to 30

(iii) Exit

---------- Cut here ----------

9. (a) Write a C++ program to sort a list of numbers in ascending order using straight insertion sort.

(b) Write a Shell program to print monthly pay-slips for employees in an organization. (Note : Assume your own data and generate accordingly).

---------- Cut here ----------

10. (a) Write a C++ program to sort a list of numbers in descending order using straight selection sort.

(b) Write a menu-driven shell program to perform the following file operations:

COPY, EDIT, RENAME, and DELETE.

DE–8842 **46**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fourth Semester

LAB VII — INTERNET PROGRAMMING

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Break up of marks :

Record Note Book : 5

Algorithm/Flow chart : 15

Program : 35

Debugging and Execution : 35

Result : 10

Total : 100

ONE question is to be given by the examiner to each candidate by lot system.

1. (a) Write a Java Program to create the frames and its controls using Applet.

   (b) Write a VB Script code to do the following :

      (i) Display the current date

      (ii) Find the difference between two dates.

      (Use dd/mm/yy format for date)

2. (a) Write a Java Program Using Applet to display the different colors and fonts.

   (b) Write a Swing program to create the tabbed panels.

---------- Cut here ----------

3. (a) Create a screen with string "web design" by using at least 20 HTML Tags.

   (b) Write a JavaScript code to create a window by using the confirm message.

---------- Cut here ----------

4. (a) Write a Java Program to create a color as the background.

   (b) Write a VB Script to do the following :

      (i) Check the given password

      (ii) Change the existing password.

---------- Cut here ----------

5. (a) Write a VB Script Code to create the calendar for the given month and year.

   (b) Write a Swing program to create buttons with

      (i) Tool tip Text

      (ii) Image

      (iii) Border.

6. (a) Write a Java Program to display dialogue menu using Applet.

(b) Write a VB Script code to do the following :

(i) Display the current date

(ii) Find the difference between two dates.

(Use dd/mm/yy format for date)

---------- Cut here ----------

7. (a) Create a web page in the format of front page of a news paper using Text links. Align text with colors.

(b) Write a Swing program to create the Tabbed Panels.

---------- Cut here ----------

8. (a) Write a JavaScript code to create the Order Form to select the house articles.

(b) Write a Swing program to create buttons with

(i) Tool tip Text

(ii) Image

(iii) Border.

---------- Cut here ----------

9. (a) Write a Java Program using Applet to display the dialogue menu.

(b) Write a Java Script code to create a color Palet and display the text in the chosen color from the Palet with proper background color.

10. (a) Write a JavaScript code to create window by using the confirm message.

(b) Write a Java program to create color as the background.

DE–8843 **47**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fourth Semester

LAB VIII – NETWORK LAB

(2005 to 2010 Calendar Year)

Time : 3 Hours Maximum : 100 Marks

Answer any ONE chosen by the candidate through lot system.

Break-up of marks

Record Note Book	: 5
Algorithm/Flow chart	: 15
Program	: 35
Debugging and Execution	: 35
Result	: 10
Total	: 100

1. (a) Write a Java program to divide two numbers using RMI client/server technique.

   (b) Write a Java program to send message from one system to another system.

---------- Cut here ----------

2. (a) Write a Java program to generate Fibonacci series of given number using RMI client/server technique.

   (b) Implement reading IP and port ID from command line and sending message to server.

3. (a) Write a Java program to check the given string is PALINDROME or NOT using RMI client/server technique.

   (b) Write a Java program to send message from one system to group of systems.

---------- Cut here ----------

4. (a) Write a Java program to find the given number is ARMSTRONG number or NOT using RMI client/server technique.

   (b) Write a Java program to ENCRYPT and DECRYPT the word at client and server respectively.

---------- Cut here ----------

5. (a) Write a Java program to subtract two numbers using RMI client/server technique.

   (b) Write a Java program for bi-directional message communication between client and server.

   (Example: client sends the message "Hi from client" and the server response will be "Hello" from Server)

---------- Cut here ----------

6. (a) Write a Java program to add two numbers using RMI client/server technique.

   (b) Write a Java program setup a chatting window server and client.

7. (a) Write a Java program to find the square-root of a number using RMI client/server.

   (b) Write a Java program to setup a Time-server and time-client.

---------- Cut here ----------

8. (a) Write a Java program to find the reverse of a given string using RMI client/server technique.

   (b) Write a Java socket program for EchoServer and EchoClient Communication.

---------- Cut here ----------

9. (a) Write a Java program to find the given number is ODD or EVEN using RMI client/server technique.

   (b) Write a Java program to read a file from the remote system.

---------- Cut here ----------

10. (a) Write a Java program to implement the client-server Communication using UDP.

    (b) Write a Java program to find the FACTORIAL of given number, using RMI client/server technique.

DE–8844 **56**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Fifth Semester

COMPUTER LAB IX — VISUAL C++

(2005 to 2010 Calender Year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Break-up marks

Record	5
Algorithm/Flow chart	15
Program	35
Debugging and Execution	35
Result	10
Total	100

1. (a) Write a VC++ win 22 application program using MFC that displays line, rectangle, polygon filled with colors.

   (b) Write a VC++ win 32 application program using MFC that displays a message “Hai”.

---------- Cut here ----------

2. (a) Write a VC++ win 32 application program using MFC that displays current mouse coordinates in status bar.

   (b) Write a VC++ win 32 application program using MFC that displays a message “Welcome”.

3. (a) Write a VC++ win 32 application program that creates two push buttons OK and CANCEL on the client area. Buttons should respond to user click over them and display appropriate message.

   (b) Write a VC++ win 32 application program using MFC that creates a list box and displays name of the states in India.

---------- Cut here ----------

4. (a) Write a VC++ win 32 application program using MFC that displays a menu. Choose the menu items using keyboard accelerator keys and displays appropriate messages for the selected command, in message box.

   (b) Write a VC++ win 32 application program using MFC that allows the user to draw pictures with the help of mouse as a free hand drawing tool.

---------- Cut here ----------

5. (a) Write a VC++ win 32 application program using MFC that displays current mouse coordinates in status bar.

   (b) Write a VC++ win 32 application program using MFC that displays rectangle, ellipse and polygon filled with colors.

---------- Cut here ----------

6. (a) Write a VC++ win 32 application program to display "Hai" whenever the user clicks the mouse button on the client area.

   (b) Write a VC++ win 32 application program using MFC that displays the status of ALT, SHIFT and SCROLL keys.

7. (a) Write a VC++ win 32 application program using MFC that fills the background of the client area with a bitmap.

   (b) Write a VC++ win 32 application program using MFC that creates a new font.

---------- Cut here ----------

8. (a) Write a VC++ win 32 application program using MFC that displays a message “welcome”.

   (b) Write a VC++ win 32 application program using MFC that displays line, polygon filled with colors.

DE–8845 **57**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. Lateral DEGREE EXAMINATION, DECEMBER 2014.

Fifth Semester

LAB X — COMPILER DESIGN

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Examiner should select and give ONE question to each candidate by lot system.

Break-Up of marks :

Record Note Book	5
Algorithm/Flow chart	15
Program	35
Debugging and Execution	35
Result	10
Total	100

1. Write a C Program for the following :

   (a) Remove spaces

   (b) Comment lines

   (c) New lines and

   (d) Produce line numbers.

---------- Cut here ----------

2. Write a C Program for the following :

   Recognize constant in the given source code.

3. Write a C Program to construct predictive parsing table for the following production rule

$S \rightarrow iC + SS1/a$

$S1 \rightarrow eS/E(E \text{ is empty})$

$C \rightarrow b$

---------- Cut here ----------

4. Write a C Program to construct of an canonical LR parsing table for the following grammar

$S \rightarrow CC$

$C \rightarrow CC/d.$

---------- Cut here ----------

5. Write a C Program to translate the expression $(a+b)*(b-c)/(a+b)$ into post fix notation and three-address code, quadruples, triples and indirect triples.

---------- Cut here ----------

6. Write a C Program for the following :

Recognize Identifier in the given source code.

---------- Cut here ----------

7. Write a C Program to construct of an SLR parsing table for the following grammar.

$E \rightarrow E + T/T$

$T \rightarrow T * F/F$

$F \rightarrow (E)/id$

8. Construct a C Program to accept a string and check whether it satisfies the following production rule (by using operator precedence grammar)

   $S \rightarrow AbB$

   $A \rightarrow aBb$

   $B \rightarrow a/Bba.$

---------- Cut here ----------

9. Write a C Program to accept a string and check whether it belongs to ambiguous grammar or not by using following production rule

   $E \rightarrow E + E/E - E/E * E/E/E/id/const/(E).$

---------- Cut here ----------

10. Write a C Program to construct the left-most derivation and right-most derivation for the following grammar.

    $S \rightarrow a /^/(T)$

    $T \rightarrow T, S/S$

    (a) (a, (a, a))

    (b) To find the leading and trailing sets.

DE–8846 **65**

DISTANCE EDUCATION

M.C.A./M.C.A. (Lateral) DEGREE EXAMINATION, DECEMBER 2014.

Sixth Semester

Computer Applications

LAB XI — MULTIMEDIA APPLICATIONS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Examiner has to choose the ONE question for each candidate by LOT system.

Each subdivision carries 50 marks.

1. (a) Write a C++ program to create a second image array model from the given image using grey-level transformation bunching technique.

1	8	4	3	6	2	5	2	8	4	6	2	5
0	3	8	3	6	5	4	0	3	8	3	8	7
0	9	8	0	5	4	8	5	9	3	7	2	9

Also draw histograms for the first and second images.

(b) Design a birth day greeting page including graphic features in Flash.

2. (a) Write a C++ program for creating an image from the given image using automatic selection of grey level for splitting technique.

1	8	4	3	6	2	5	2	8	4	6	2	5
0	3	8	3	6	5	4	0	3	8	3	8	7
0	9	8	0	5	4	8	5	9	3	7	2	9

(b) Design an animated cartoon picture.

---------- Cut here ----------

3. (a) Write a C++ program for setting up a pattern for multi-grey-level images or bilevel display using limbs algorithm with the following $2 * 2$ matrix.

$$\begin{matrix} 0 & 2 \\ 3 & 1 \end{matrix}$$

Also print the dot patterns produced on a 16 grey-level system for a pixel.

(b) Prepare a multimedia presentation shows the features of a software package developed by you.

---------- Cut here ----------

4. (a) Write a C++ program to enhance an image using median filter.

(b) Prepare a multimedia presentation showing the features of the courses offered in a College.

---------- Cut here ----------

5. (a) Write a C++ program to perform image subtraction operation.

(b) Write a C++ program for grey level transformation techniques :

(i) Thresholding

(ii) Bunching.

6. (a) Write a C++ program for convolution using given template :

Image					Template	
1	1	3	3	4	1	0
1	1	4	4	3	0	1
2	1	3	3	3		
1	1	1	4	4		

(b) Write a C++ program for 2 dimensional image animation using computer graphics.

---------- Cut here ----------

7. (a) Write a C++ program to find the edges of the following image using Sobel edge detection procedure.

(b) Write a C++ program that smoothens an image in the transform domain.

---------- Cut here ----------

8. (a) Write a C++ program to compress the image compression using contour coding technique.

Given image :

1	1	1	1	1	1	1	1	2	2	2	2	2	2	2
1	1	1	1	1	1	1	2	2	2	2	2	2	2	2
1	1	1	1	1	1	2	2	2	2	2	2	2	2	2
1	1	1	1	1	1	1	2	2	2	2	2	2	2	2
1	1	1	1	1	1	1	1	1	1	1	2	2	2	2

(b) Do the image animation using Flash.

9. (a) Write a C++ program to compress an image using contour coding technique.

(b) Design a sales advertisement for a product using Flash.

---------- Cut here ----------

10. (a) Write a C++ program to find the best grey level for splitting the following image :

4	5	1	7	4	2	6	3	4	2	7	4	6	7
3	7	8	2	7	6	5	4	3	3	7	2	4	3
4	5	9	9	9	7	8	7	6	5	5	4	1	2
0	0	0	3	4	2	2	3	1	1	4	6	6	5
7	3	2	4	5	1	6	2	3	4	5	6	7	9

Using automatic selection for splitting technique.

(b) Design an invitation for your college day function including various graphic features in flash.